Public sitemap & interaction model

Visitor-facing information architecture, URL targets and primary task flow

Solid lines: page composition

Dashed outline: retained direct route

Blue arrows: interaction / progression

PUBLIC HOME

/ - One continuous page with embedded catalogue and engineering workspaces

FA / RTL | EN / LTR

01 Company & context

`/ | #about | #industries`

- Hero and quick product search
- Capabilities and company profile
- Industries and applications
- Why choose us / technical support

02 Product catalogue

`/#catalog`

- Search: code, size, application
- Family and dimension filters
- Product results and selection
- Open product information panel

03 Engineering workspace

`/#engineering`

- Product selection / input context
- Life and equivalent load
- Basic rating life and fit limits
- Technical comparison
- Unit conversion

04 Contact & inquiry

`/#contact`

- Company contact information
- Technical inquiry form
- Submission feedback

RETAINED DIRECT ROUTES

`/catalog` `/engineering`

`/product/:slug`

PRODUCT INFORMATION PANEL

Identity / specifications / images / illustrative 3D / documents

`Shareable selection: /?item=:slug#catalog`

SHARED SITE SHELL

Header / section navigation / footer

Language and reading direction

PRIMARY VISITOR FLOW

01 / DISCOVER Search and filter → **02 / INSPECT** Review product details → **03 / EVALUATE** Use engineering tools → **04 / REFERENCE** Read documents → **05 / CONNECT** Send a technical inquiry

Recommended walkthrough; users may jump between sections and tools.

DESIGN CONTRACT

Preserve bilingual RTL/LTR layouts, technical data hierarchy, product context and clear interaction states.

Design review coverage: keyboard/focus, loading, empty/error states, mobile reflow and reduced-motion behavior.

Source: app/Router.tsx; Home.tsx; HomeSections.tsx; Engineering.tsx; lib/public-links.ts. Sections are grouped for readability.

Administration & data architecture

Application boundaries, content ownership and the operational inquiry loop

Two-way arrows: requests and responses, or storage reads and writes. Module boundaries are shown explicitly.

PUBLIC INTERFACE

React + TypeScript

Company / catalogue / contact

Client-side engineering tools

Three.js illustrative mechanical views

Reads public content; submits inquiries.

ADMIN CONTROL CENTER

`/admin | /admin/`

Authenticated access / role-based permissions

- Overview
- Products
- Media
- Company
- Inquiries
- Website editor
- Header
- SEO
- System & security
- Audit logs

Public content responses
Inquiry submissions

Authenticated reads / writes
Uploads and status updates

APPLICATION SERVER

Node + Express API

Public data and inquiries / protected administration / validation

Server sessions / media handling / audit events / backup endpoints

Returns saved content and operation results to the relevant interface.

Read / write structured records

Store / serve file bytes

SQLITE / AUTHORITATIVE SAVED DATA

Products / website content / company settings / inquiries

Media metadata / accounts and sessions / audit records

MEDIA STORAGE / UPLOADED FILES

Image and PDF bytes are stored separately from SQLite.

Product and content records refer to media resources.

INQUIRY OPERATIONS

Visitor submits → **NEW** → **REVIEWED** → **CONTACTED** → **CLOSED**

Staff change inquiry status. Engineering formulas remain in application code; routine content is managed in the CMS.

Source: README.md; components/admin/Admin.tsx; platform components. Deployment topology and entity-level schema are outside this view.